

MOHAMED AYMAN SALEM

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PROFESSIONAL SUMMARY

AI-focused Computer Science graduate with hands-on experience building and deploying end-to-end ML applications. Proficient in Python, TensorFlow, FastAPI, and OpenCV. Expanding expertise in Microsoft Power Platform to automate workflows and integrate AI into business solutions. Passionate about LLMs, RAG systems, computer vision, and scalable API design.

EDUCATION

B.Sc. Computer Science — Suez Canal University, Faculty of Computers & Information

Grade: Very Good | 2020 – 2024

TECHNICAL SKILLS

Languages: Python (Async, OOP), C++, C

AI / ML & Computer Vision: TensorFlow, Keras, OpenCV, DeepFace, Tesseract OCR, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Backend & APIs: FastAPI, Flask, RESTful APIs, Gemini API, Piston API, LangChain, FAISS

Microsoft Power Platform: Power Apps, Power Automate, Power BI, Power Pages, Copilot Studio

DevOps & Tools: Docker, Git, GitHub, Linux, Microsoft Excel

EXPERIENCE

Python & AI Instructor — TROSC Team

2022 – 2024

- Conducted hands-on training sessions in Python and AI for groups of 50+ students
- Delivered practical curriculum covering machine learning, computer vision, and real-world project development

Data Science Intern — El Masriya Co.

2023

- Attended a structured summer internship focused on Data Science fundamentals, including data analysis, visualization, and ML concepts

VOLUNTEER

Team Leader — TROSC Team

2 Years

- Led a tech-focused team bridging the gap between academia and industry through bootcamps, training tracks, and professional events

PROJECTS

Personal AI Agent — RAG-Powered Portfolio Assistant | GitHub

LangChain | FastAPI | Gemini API | FAISS | Docker | Hugging Face Spaces

- Built a production-ready RAG pipeline ingesting CV, LinkedIn, and GitHub data into a FAISS vector store
- Implemented per-session conversation memory, IP-based rate limiting, and secure admin endpoints for live CV management
- Deployed on Hugging Face Spaces via Docker with Git LFS; integrated live into portfolio at poortfllo.netlify.app

Blockchain-Based Secure Voting System | GitHub

Python | Flask | OpenCV | DeepFace | TensorFlow | Tesseract-OCR

- Built AI facial recognition with DeepFace for voter identity verification and anti-spoofing via deep learning
- Designed OCR-based ID verification pipeline with database cross-checking and multi-layer blockchain security

Smart Learning Platform — AI-Powered System | GitHub

FastAPI | Gemini API | Piston API | Async Python

- Developed personalized learning paths with adaptive MCQ and live coding assessments
- Built context-aware AI tutor with AI-driven quiz generation, automated grading, and multi-language code evaluation
- Improved reliability via retry mechanisms and multi-model fallback strategies

Skin Lesion Classification — MobileNetV3 | GitHub

TensorFlow | Keras | MobileNetV3Small | scikit-learn | TFLite

- Classified 7 skin diseases using MobileNetV3Small pre-trained on ImageNet; achieved 81.6% test accuracy and 0.984 AUC
- Applied medical-grade augmentation and class-weighted loss — improved Melanoma recall by ~18%
- Exported to TFLite (float16, 4 MB, <1ms inference) for Android/iOS deployment

CERTIFICATIONS & COURSES

- ML Specialist — DeepLearning.AI
- Microsoft Power Platform Fundamentals — Coursera
- Introduction to Git & GitHub — Google
- Get Started with Python — Google

ADDITIONAL

Languages: Arabic (Native), English (Professional) | **Competitive Programming:** ECPC Problem Solving Contestant (2021–2024)